

MorphGrower: A Synchronized Layer-by-layer Growing Approach for Plausible Neuronal Morphology Generation

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Code: <https://github.com/Thinklab-SJTU/MorphGrower>

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Method Overview: What We Propose

- Layer-by-layer generation; sibling branches decoded in synchronized pairs.
- Each layer conditioned on prior structure: local path + global previous layers.
- CVAE with vMF latent; branch LSTMs; global forest GNN+GRU; local EMA.
- Soma layer: provided or via unconditional VAE (†).



Architecture diagram (Fig. 2): CVAE with branch LSTMs, global forest GNN+GRU, local EMA, paired decoder

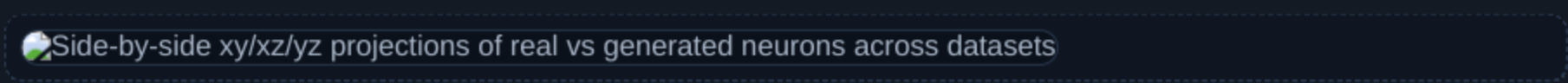
Architecture (Fig. 2): branch LSTMs + global forest encoder (GNN+GRU) + local EMA feed a CVAE (vMF latent); decoder outputs sibling branches jointly.

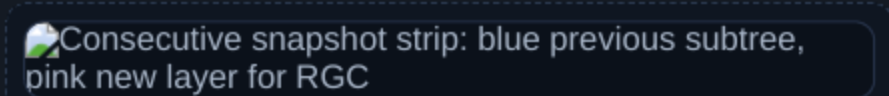


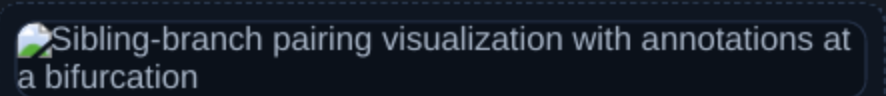
Generation snapshot: reference layer, generated subtree, newly synthesized layer

Layer-wise inference: condition on generated subtree and reference layer to synthesize the next layer.

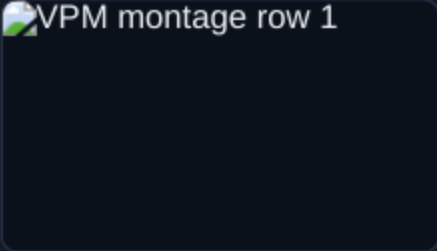
Qualitative Examples: Realistic, Valid Morphologies

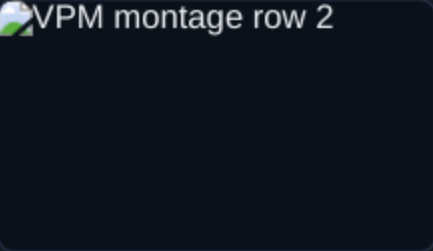


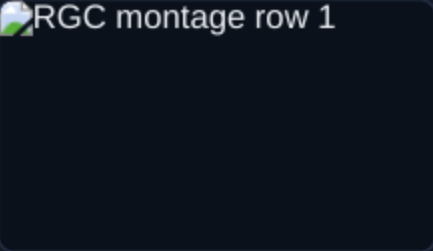


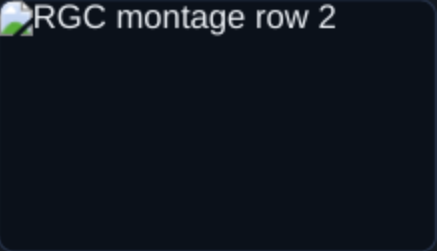


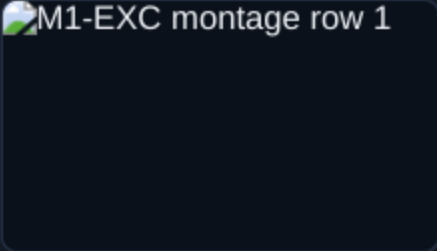
Qualitative Montages Across Neuron Types

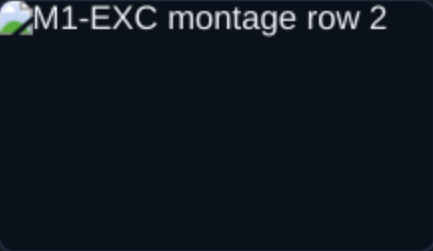


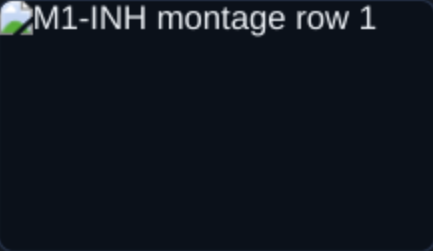


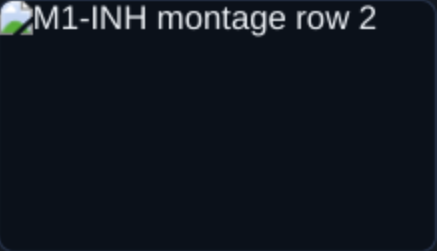




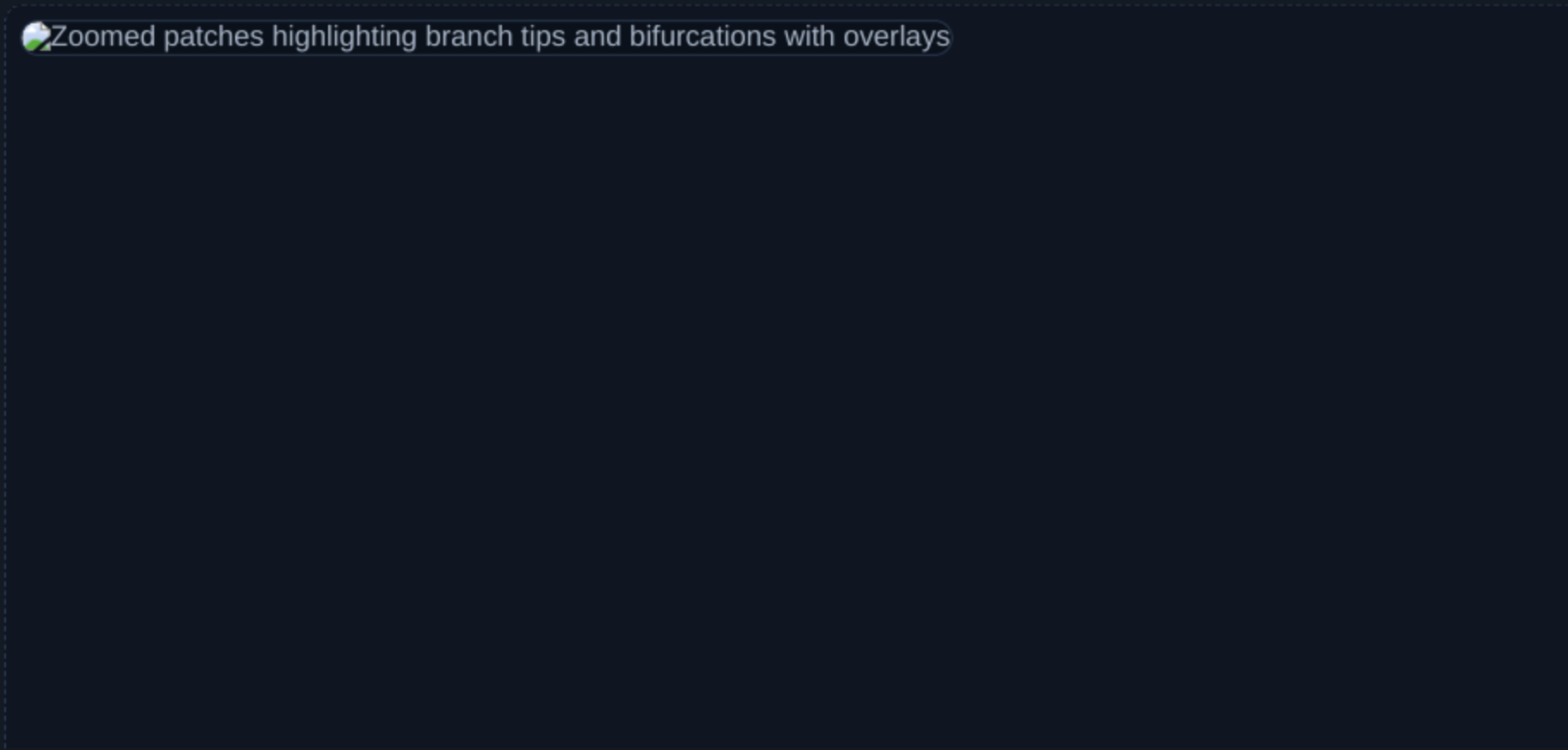








Zoomed Qualitative Details: Tips and Bifurcations



Takeaway & Contact

- Synchronized, layer-wise paired-branch growth yields valid, realistic morphologies conditioned on prior structure.

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